

UNIT - 4

Hypothesis Testing

Hypothesis is an assumption or statement or population parameter

Hypothesis testing is a formal procedure for investigating ideas about the world using statistics

steps in hypothesis testing

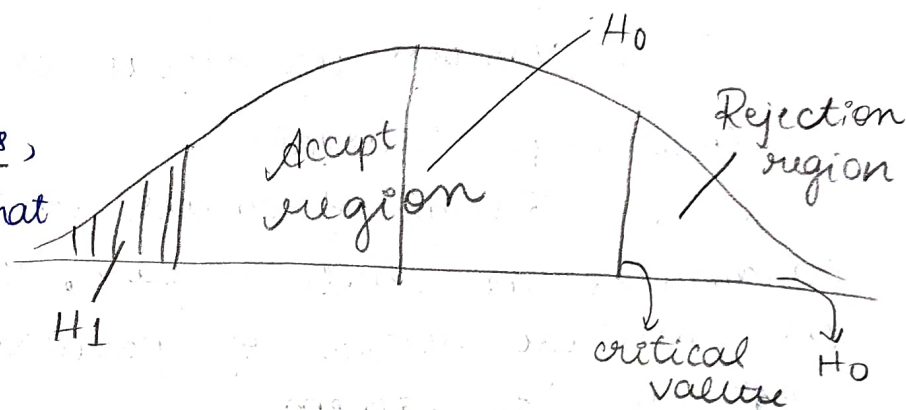
1. State your research hypothesis as NULL hypothesis H_0 and Alternative hypothesis H_1
2. Collect data, test the hypothesis
3. Perform appropriate statistical tests
4. Decide whether to reject or fail to reject null hypothesis.

H_0 is null hypothesis, an initial hypothesis that predicts relationship between variables

H_1 Alternative hypothesis, it is a prediction of no relationship between variables.

eg: ~~H_0~~ H_0 : A ~~is~~ taller than B

H_1 : A is not taller than B



Direction or Type of Tail

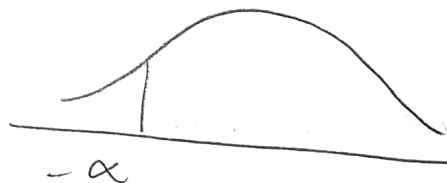
1. Double tailed test



$$H_0 : \mu = \mu_0$$

$$H_1 : \mu \neq \mu_0$$

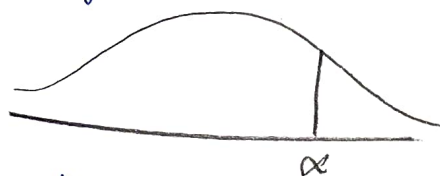
2. Left Tail Test



$$H_0 : \mu = \mu_0$$

$$H_1 : \mu < \mu_0$$

3. Right tail test



$$H_0 : \mu = \mu_0$$

$$H_1 : \mu > \mu_0$$

$\mu \rightarrow$ population
 $\mu_0 \rightarrow$ sample

Simple and Composite Hypothesis

When hypothesis specifies exact value of the parameter then it is a simple hypothesis.

When the hypothesis specifies range of values then it is composite hypothesis.

eg: Simple hypothesis: The age

The various techniques employed are:

Confidence intervals:

Range of values within which population parameter is likely to fall. 95% confidence interval for average height of population might be 165 cm to 175 cm.

P-value in Statistical hypothesis test

- The p-value is the probability value with the assumption that null hypothesis is true.
- P-values are generally expressed as decimals and it is used in hypothesis testing to help support or reject null hypothesis.
- Smaller the p-value stronger the evidence to reject null hypothesis.

Student t-test

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n-1}}$$

$\bar{x}$ → sample mean

μ → population mean

s → sample std deviation

n → size of the sample

Significance level at 10%, 5% or 1%

test score in accept zone means failed to reject null hypothesis

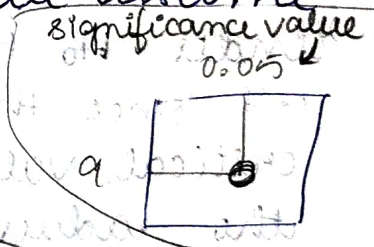
~~accept~~ p-value is a cut-off value within accept zone and reject zone.

1. 10 items were taken at random from an automatic drilling machine, the mean weight of 10 items is 11.8 and the std deviation is 0.15. Does the sample mean differs significantly from the indentent weight of 12. Calculate the population mean & sample mean.

as (increase or decrease keyword not mentioned it is 2 tailed test) $FP.1 = \frac{1 - 0.05}{2} = 0.475$

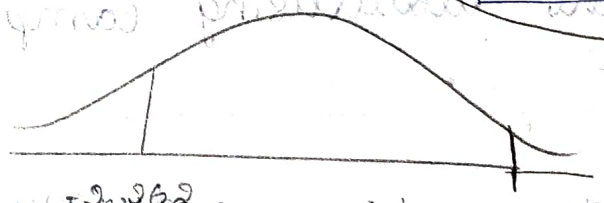
Given, $\bar{x} = 11.8$ $s = 0.15$
 $\mu = 12$ $n = 10$

Note: When significant level not given we assume it to be 0.5



$$t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}} = -4$$

$$t_{0.5} = 2.262$$



at 9° of freedom, the level of significance is 0.05 level $t_{0.05}$

The sample mean differs at $\mu = 12$.

H_0 is rejected or failed to accept H_0

2. The mean weekly sales of a chocolate in stores was 146.3 bars/store. After an advertising campaign, the mean weekly sales of 22 stores for a typical week increased to 153.7 and showed a std deviation of 17.2.

was the advertising campaign successful.

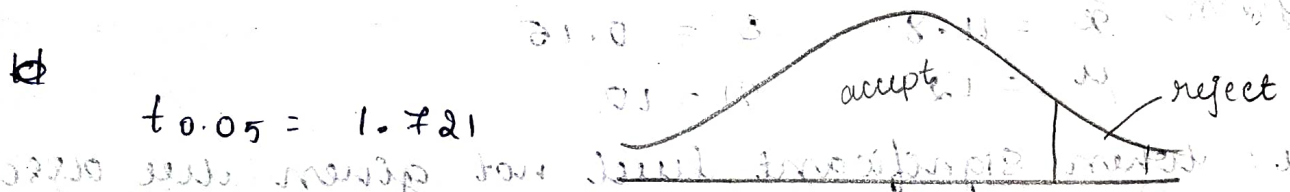
[successful means increase, therefore single tailed, right as increase]

mean $\bar{x} = 153.7$ $n = 22$ $s = 17.2$

$\mu = 146.3$ $H_0: \mu = \mu_0$ $H_1: \mu > \mu_0$

under H_0 .

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}} = \frac{153.7 - 146.3}{17.2 / \sqrt{22-1}} = 1.97$$

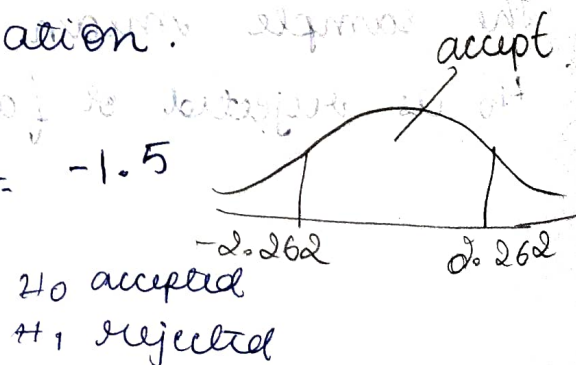


Under H_0 the $t = 1.97$ and the tabulated value is 1.7. Since the calculated value lies outside the critical value and favorable value to H_1 and the advertising campaign was successful.

2. A machine is designed to produce insulating washers for electrical wires of average thickness of 0.025 cm. A random sample of 10 washers was found to have an average thickness of 0.024 cm with std deviation 0.002 cm. Test the significance of the deviation.

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}} = \frac{0.024 - 0.025}{0.002 / \sqrt{10-1}} = -1.5$$

$t_{0.05} = 2.262$



H_0 accepted

H_1 rejected

t-test for difference of means.

$$t = \frac{\bar{x} - \bar{y}}{s \times \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$\bar{x} = \frac{1}{n_1} \sum x$$

$$s = \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$s^2 = \frac{1}{n_1 + n_2 - 2} \left[\sum (x - \bar{x})^2 + \sum (y - \bar{y})^2 \right]$$

Assumptions for difference of means.

1. The parent population from which samples have been brought are normally distributed.
2. Two samples are random and independent of each other.
3. ~~$\sigma_1^2 = \sigma_2^2$~~ $\sigma_1^2 = \sigma_2^2 = \sigma^2$ The population variances are equal.

$$\sum (x - \bar{x})^2 = \sum d^2 - \frac{(\sum d)^2}{n_1}$$

$$\sum (y - \bar{y})^2 = \sum d^2 - \frac{(\sum d)^2}{n_2}$$

1. The content in milli grams of 2 samples of a product were found to be as follows

Sample A 24, 27, 26, 21, 25

double tailed

Sample B 27, 30, 28, 31, 22, 36

- a) Can it be said 2 samples come from normal population having same mean.

Assumed mean for sample A 25 for B is 30.

~~Assume~~ given critical value 2.26 Given degree of freedom is 9

$$\Rightarrow \bar{x} = \frac{\sum x}{n_1}$$

X	$d_1 = x - A$	d_1^2
24	-1	1
27	2	4
26	1	1
21	-4	16
25	0	0

Y	$d_2 = y - A$	d_2^2
27	2	4
30	5	25
28	3	9
31	6	36
22	-8	64
36	11	121

sample A			sample B		
X	$d_1 = X - A$	d_1^2	Y	$d_2 = Y - B$	d_2^2
24	-1	1	27	-3	9
27	2	4	30	0	0
26	1	1	28	-2	4
21	-4	16	31	1	1
25	0	0	22	-8	64
			36	6	36
	-2	22		-6	114

$$\bar{X} = A + \frac{\sum d_1}{n_1} = 25 + \frac{(-2)}{5} = 24.6$$

$$\bar{Y} = B + \frac{\sum d_2}{n_2} = 30 + \frac{-6}{6} = 29$$

$$\sum (x - \bar{x})^2 = \sum d_1^2 - \frac{(\sum d_1)^2}{n_1} = 22 - \frac{(-2)^2}{5} = 21.2$$

$$\sum (y - \bar{y})^2 = \sum d_2^2 - \frac{(\sum d_2)^2}{n_2} = 114 - \frac{(-6)^2}{6} = 108$$

$$S^2 = \frac{1}{n_1 + n_2 - 2} \left[\sum (x - \bar{x})^2 + \sum (y - \bar{y})^2 \right] = \frac{1}{5 + 6 - 2} [21.2 + 108]$$

$$14.3556$$

$$S = 3.79$$

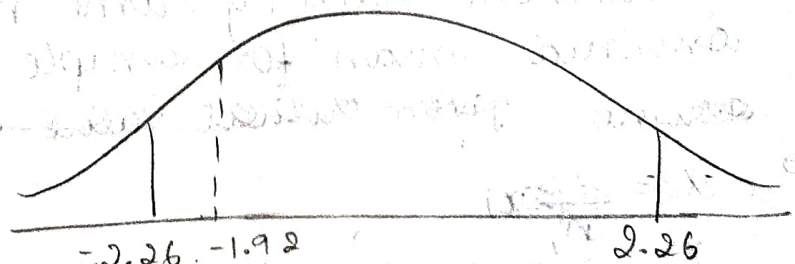
$$t = \frac{\bar{x} - \bar{y}}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{24.6 - 29}{3.79 \sqrt{\frac{1}{5} + \frac{1}{6}}} = -1.92$$

df = 9

$$t_{0.05} = 2.26$$

H_0 accepted $\mu_A = \mu_B$

H_1 rejected $\mu_A \neq \mu_B$



$\therefore$ the two samples are from the same population.

2. A group of 5 patients treated with medicine A are 42, 39, 48, 60 and 41. Second group of 7 patients from the same hospital treated with medicine B are 38, 42, 56, 64, 68, 69 and 62. Do you agree with the claim that medicine B increases the weight significantly given that value of t at 5% level of significance for 10° of freedom is 2.228 [Assumed mean for A 46, for B 57]

Medicine A			Medicine B		
X	$d_1 = X - A$	d_1^2	Y	$d_2 = Y - B$	d_2^2
42	-4	16	38	-19	361
39	-7	49	42	-15	225
48	2	4	56	-1	1
60	14	196	64	7	49
41	-5	25	68	11	121
			69	12	144
			62	5	25
	0	290		0	926

$$\bar{X} = A + \frac{\sum d_1}{n_1}$$

$$= 46 + \frac{0}{5} = 46$$

$$\bar{Y} = B + \frac{\sum d_2}{n_2}$$

$$= 57 + \frac{0}{7} = 57$$

$$(\sum (x - \bar{x})^2) = \sum d_1^2 - \frac{(\sum d_1)^2}{n_1} = 290 - \frac{0^2}{5} = 290$$

$$(\sum (y - \bar{y})^2) = \sum d_2^2 - \frac{(\sum d_2)^2}{n_2} = 926 - \frac{0^2}{7} = 926$$

$$S^2 = \frac{1}{n_1 + n_2 - 2} [\sum (x - \bar{x})^2 + \sum (y - \bar{y})^2] = \frac{1}{5 + 7 - 2} [290 + 926] = 121.6$$

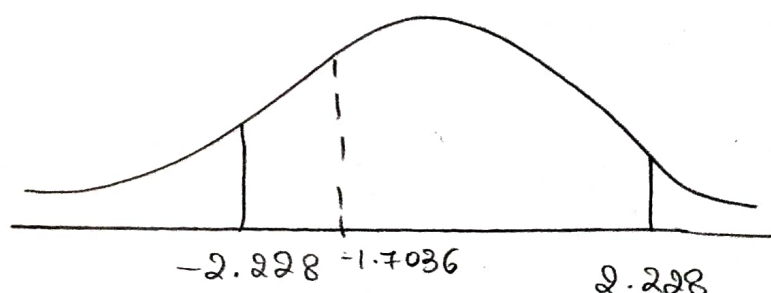
$$S = \sqrt{121.6} = 11.027$$

$$t = \frac{\bar{x} - \bar{y}}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{46 - 57}{11.027 \sqrt{\frac{1}{5} + \frac{1}{7}}} = -1.7036$$

$$t_{0.05} = 2.228$$

$$H_0: \mu_A = \mu_B$$

$$H_1: \mu_A < \mu_B$$



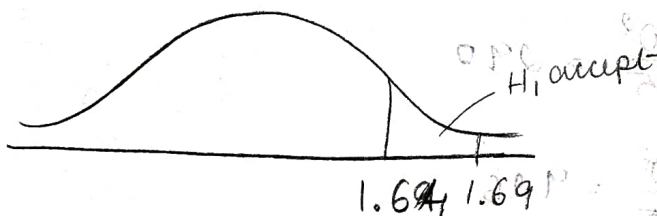
H_0 accept, H_1 reject
 H_0 is accepted, there is no significant difference between the medicines A and B

3. A random sample of 20 workers of state A was found to have average earning of ₹44 with a sample variance 900. Another sample of 20 workers from state B was found to earn an average of ₹30 per day, with a sample variance 400. Test whether the workers in state A are earning more than those in state B. Given that the critical value to be 1.64.

Note: When variance is given $S^2 = \left(\frac{n_1 S_A^2 + n_2 S_B^2}{n_1 + n_2 - 2} \right)$

Solⁿ: $n_1 = 20$ $n_2 = 20$ $S^2 = \left(\frac{n_1 S_A^2 + n_2 S_B^2}{n_1 + n_2 - 2} \right) = \frac{20(900) + 20(400)}{20 + 20 - 2}$
 $S_A^2 = 900$ $S_B^2 = 400$
 $\bar{x} = 44$ $\bar{y} = 30$
 $S^2 = 684.21$
 $S = 26.157$

$$t = \frac{\bar{x} - \bar{y}}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{44 - 30}{26.157 \sqrt{\frac{1}{20} + \frac{1}{20}}} = 1.6925$$



$$H_0 : \mu_A = \mu_B$$

$$H_1 : \mu_A > \mu_B$$

~~H_0 accepted~~ ~~H_0 accepted~~

H_0 rejected H_1 accepted

5% confidence workers in state A is earning more than state B

Paired t-test for difference of means

$$t = \frac{\bar{d}}{s/\sqrt{n}} = \frac{\bar{d}}{\sqrt{s^2/n}}$$

$$d = x - y$$

$$\bar{d} = \frac{1}{n} \sum d$$

$$s^2 = \frac{1}{n-1} \sum (d - \bar{d})^2$$

$$s^2 = \frac{1}{n-1} \left[\sum d^2 - \frac{(\sum d)^2}{n} \right]$$

1. An IQ test was given to 5 people before and after they were trained. The results are given below

Candidate	IQ before training	IQ after training
I	110	120
II	120	118
III	123	125
IV	132	136
V	125	121

Test whether there is any change in IQ after training. Given that the critical value to be 4.6 at 1% level of significance.

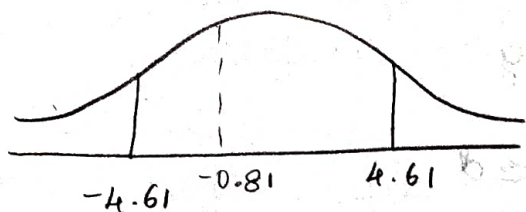
candidate	x before	y after	$d = x - y$	d^2	$\bar{d} = \frac{\sum d}{n}$
I	110	120	-10	100	
II	120	118	2	4	
III	123	125	-2	4	
IV	132	136	-4	16	
V	125	121	+4	16	
			-10	140	

$$\bar{d} = \frac{1}{n} \sum d = \frac{1}{5} (-10) = -2$$

$$s^2 = \frac{1}{n-1} \left[\sum d^2 - \frac{(\sum d)^2}{n} \right] = \frac{1}{4} \left[140 - \frac{(-10)^2}{5} \right] = 30$$

$$s = 5.477$$

$$t = \frac{\bar{d}}{\sqrt{s^2/n}} = \frac{-2}{\sqrt{30/5}} = -0.81$$



~~H_0~~ $H_0: \mu_0 = \mu$

$H_1: \mu_0 \neq \mu$

H_0 is accepted H_1 rejected.

There is no significant change in IQ after training

Non Parametric test and Parametric test

Parametric test is a test whose model specifies certain conditions about parameters of the population from which samples have been drawn.

Non parametric test is a test that doesn't depend on a ~~test~~ a particular form from which samples are drawn.

Mann-Whitney Test

Compares population ~~medians~~ ^{median} of x and population median of y . It compares 2 independent samples by testing hypothesis on 2 population medians. Let $x_1, x_2, \dots, x_n$ then $y_1, y_2, \dots, y_n$ be independent random samples of size n_1 and n_2 from 2 populations.

$$n_x \text{ \& } n_y$$

v-test

$$V_x = T_x - \frac{n_1(n_1+1)}{2}$$

$$V_y = T_y - \frac{n_2(n_2+1)}{2}$$

T_x is the sum of ranks of first sample

T_y is the sum of ranks of 2nd sample.

Steps to be followed:

Step 1: Set up hypothesis

$$H_0: \eta_x \geq \eta_y \text{ or}$$

$$\eta_x \leq \eta_y \text{ or}$$

$$\eta_x = \eta_y$$

$$\eta_x \neq \eta_y \text{ [2 tailed]}$$

$$H_1: \eta_x < \eta_y \text{ [Left tailed]}$$

$$\eta_x > \eta_y \text{ [Right tailed]}$$

Step 2: Set level of significance.

Step 3: Combine the observations of both samples to obtain single sample and arrange observations of combined samples in ascending order.. then assign ranks to combined sample T_x, T_y, U_x and U_y .

Step 4: Compute test statistics $U = \text{Smaller } (U_x, U_y)$

Step 5: Critical values and critical regions

Step 6: Decision: Reject H_0 if U lies in critical region else fail to reject H_0

Step 7: Conclusion.

Assumptions of non parametric tests that are applicable to Mann-Whitney test

1. Observations need to be independent
2. Dependent variable that is taken in research needs to display ~~dependence~~ continuity
3. The test is used when two samples, subgroups are independent and this data is ordinary.

In case of data, is interval or ratio the sample is converted to rank order. ~~Sample subgroup~~

4. Sample sub groups need to be independent & not co-related.

1. A school concellor wanted to study whether gender differences exist in students with regard to mathematical abilities. Test the hypothesis using Mann Whitney U-test. Use 0.05 significance.

Male	6	13	7	11	5
female	10	16	12	14	17

Solⁿ $H_0: \eta_x = \eta_y$ [no significant change]
 [Population median $\eta_x =$ population median η_y]
 $H_1: \eta_x \neq \eta_y$

$$\alpha = 0.05$$

T_x : Male Male (A)	6	13	7	11	5
Ranks	2	7 3	3	4 5	1
T_y : Female (B)	10	16	12	14	17
Ranks	4	9	6 8	8	10

$$T_x = 18 \text{ [sum of ranks]}$$

$$T_y = 37 \text{ [sum of ranks]}$$

$$U_x = T_x - \frac{n_1(n_1+1)}{2} = 18 - \frac{5(5+1)}{2} = 3$$

$$U_y = T_y - \frac{n_2(n_2+1)}{2} = 37 - \frac{5(5+1)}{2} = 22$$

$$U = \text{Smaller}(U_x, U_y) = 3$$

Incomplete

find from table & infer abt hypothesis

Kruskal's vallig test & H-test

It is a extension of Mann Whitney test

step 1: ~~define~~ define null and alternative hypothesis

step 2: Rank the sample observation in combined series,
compute t_i which is sum of the sample.

step 3: compute test statistics

step 4: conclusion.

$$H = \frac{12}{n(n+1)} \sum_{i=1}^k \frac{T_i^2}{n_i} - 3(n+1)$$

1. A companies trainees are randomly divided in 3 groups of 10 each and are given a course in management skills by 3 different methods. At the end of the training period they are given a test and their scores are as follows

method	A	B	C
data	99	83	89
	64	102	98
	101	125	56
	85	61	105
	99	91	87
	88	96	90
	97	94	87
	95	89	101
	90	93	76
	100	75	89

Use Kruskal vallig test to determine at 5% level of significance if 3 methods are equally effective

Solⁿ:

Sample	C	B	A	B	E	A	B	A	C	C
Observation	56	64	64	75	76	77	83	85	87	87
Rank	1	2	3	4	5	6	7	8	9	10
									9.5	9.5

Sample	A	B	C	C	A	B	B	B	A
Observation	88	89	89	90	90	91	93	94	95
Rank	11	12.5	13	14.5	15	16	17	18	

A	B	C	C	C	A	B	B	B	
88	89	89	89	90	90	91	93	94	95
11	13	13	13	15.5	15.5	17	18	19	20

	A	A	A	C	B	C	
97	98	99	100	101	101	102	105
22	23	24	25	26.5	26.5	28	29

$$T_1 = 24 + 3 + 26.5 + 8 + 6 + 11 + 22 + 20 + 15.5 + 25 = 161$$

$$T_2 = 159$$

$$T_3 = 145$$

$$\sum \frac{T_i^2}{n} = \frac{T_1^2}{n} + \frac{T_2^2}{n} + \frac{T_3^2}{n} = \frac{161^2 + 159^2 + 145^2}{10} = 7222.7$$

$$H = \frac{12}{106}$$

$$H = \frac{12}{n(n+1)} \sum_{i=1}^k \frac{T_i^2}{n} - 3(n+1)$$

$$n = 10 + 10 + 10 = 30$$

$$\sum \frac{T_i^2}{n}$$

$$H =$$

$$H = \frac{12}{30(30+1)} \left(\frac{161^2}{10} + \frac{159^2}{10} + \frac{145^2}{10} \right) - 3(30+1)$$

$$= 0.1961$$

Variation in ANOVA

1. Between groups

$$\sum n_i (\bar{x}_i - \bar{x})^2$$

2. within groups

$$\sum \sum (x_{ij} - \bar{x}_i)^2$$

3. Total

$$\sum \sum (x_{ij} - \bar{x})^2$$

df $\rightarrow$ degree of freedom

SS $\rightarrow$ sum of squares

mss $\rightarrow$ mean sum of squares.

one way ANOVA

Source of variations	df degree of freedom	SS	mss = $\frac{SS}{df}$	F-ratio

Steps to solve ANOVA

1) compute correlation factor, $c = g^2 / n$

2. compute SS, total, SS b/w grps

SS total = $\sum \sum x_{ij}^2 - c$

$$SS \text{ b/w grp} = \sum_{i=1}^g \frac{\sum x_i^2}{n_i} - c$$

$$SS \text{ within grp} = SS \text{ total} - SS \text{ b/w grp}$$

Problems

1. In an experiment to determine the effect of nutrition on attention span of elementary school students, a group of 15 students were randomly assigned to each of 3 meal plans: no breakfast, light and full breakfast. Their attention plans were recorded during morning. Construct an analysis of variance table for this

No breakfast	8	7	9	13	10	Total 47
light breakfast	14	16	12	17	11	70
full breakfast	10	12	16	15	12	65
Grand total						182
G =						182

$$C = \frac{G^2}{N} = \frac{182^2}{15} = 2208.26\bar{6}$$

one way anova

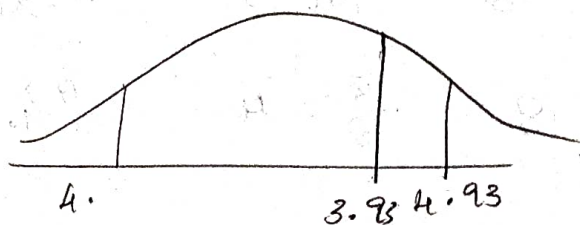
Source of Variation	df	SS	MSS = $\frac{SS}{df}$	F-ratio
Between grp	3-1 = 2	58.54	29.27	$\frac{29.27}{5.93} = 4.9$
within grp grp	14-2 12	$129.74 - 58.54$ 71.2	5.94	
Total	15-1 = 14	129.74		

$$SS \text{ total} = \sum \sum x_{ij}^2 - C = 8^2 + 7^2 + 9^2 + 13^2 + 10^2 + 14^2 + 16^2 + 12^2 + 17^2 + 11^2 + 10^2 + 12^2 + 16^2 + 15^2 + 12^2 - 2208.26\bar{6} = 129.74$$

$$SS \text{ b/w grp} = \sum_{i=1}^k \frac{\bar{x}_i^2}{n_i} - C = \frac{47^2}{5} + \frac{70^2}{5} + \frac{65^2}{5} - 2208.26\bar{6} = 58.54$$

Given that $F(2, 12)$ where 2 is degree of freedom, 12 will be within grp ~~3-2~~
critical value = 3.89 and obtained value 4.93

check
once



H_0 - Rejected

Two way Anova:

Determines how two factors impact response variable as dependent and independent variables.

	df	SS	MSS
Between row	$r - 1$	SSR	$\frac{SSR}{(r - 1)}$
Between col	$c - 1$	SSC	$\frac{SSC}{c - 1}$
within error	$(r - 1)(c - 1)$	SSE	$\frac{SSE}{df}$

Problems

1.

Fertilizers plot	yield				Total.
	A	B	C	D	
Nitrogen	6	4	8	6	24
Potash	7	6	6	9	28
phosphates	8	5	10	9	32

$$21 \quad 15 \quad 24 \quad 24 = 84$$

$$C = \frac{G^2}{N} = \frac{84^2}{72} = 588$$

	Df	SS	MSS	F-ratio
Between Row	2-1 2 2	8	$\frac{8}{2} = 4$	$4/1.6 = 2.5$
Between col	4-1 3	18	$\frac{18}{3} = 6$	$6/1.6 = 3.75$
within error	11-2-3 6 6	36-18-8 10	$\frac{10}{6} = 1.67$	
total	12-2-3 10 12-1 11	36		

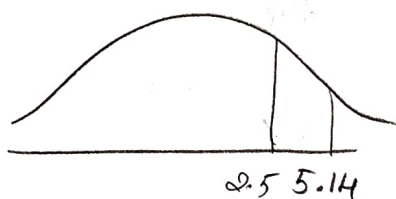
~~SSR =~~ ~~$\frac{24^2}{4} + \frac{28^2}{4} + \frac{32^2}{4} = 588$~~

$SSR = \frac{24^2}{4} + \frac{28^2}{4} + \frac{32^2}{4} = 588$

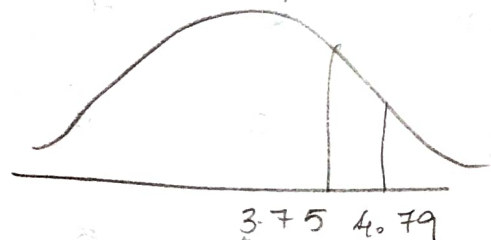
$SSC = \frac{21^2}{3} + \frac{15^2}{3} + \frac{24^2}{3} + \frac{24^2}{3} = 588$

$SS \text{ of total} = 6^2 + 4^2 + \dots + 9^2 - 588 = 36$

Critical value of $F(2, 6) = 5.14$
 $F(3, 6) = 4.79$



H_0 accept



H_0 accept